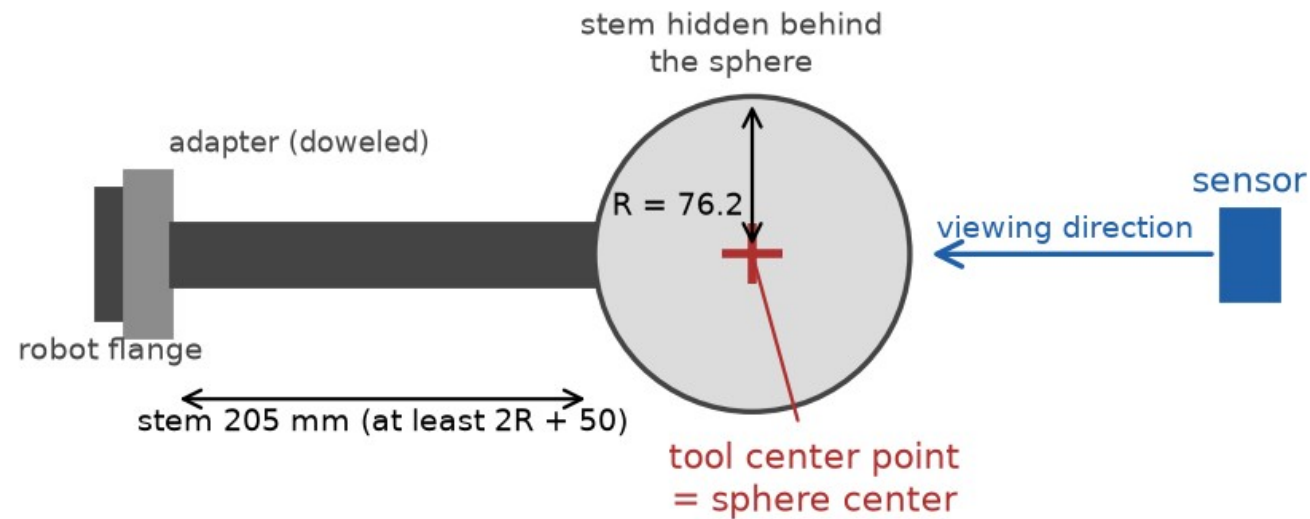


Stage-1 Calibration: Procurement and Build

The sphere, board and fixtures needed for the depth-sensor calibration captures



One sphere and a board carry the known geometry



Sphere on its stem, side view: the stem points away from the sensor so that the sphere hides it.



Sphere B

152.4 mm diameter, turned aluminum, matte; the only sphere



Curvature sweep

Nine stations from 300 to 1,100 mm, each about 1.19 times farther than the last



Board

200 x 150 mm, at least 6 mm thick, flat to 0.05 mm after painting; tilted 0, 20 and 40 degrees

What must be built

Item	Quantity	Drawing
Sphere B with mounting interface	1	SC1-04 (interface detail)
Sphere-flange mount: adapter plate and stem	1 + 1	SC1-02, SC1-04
Flat target (board)	1	SC1-05 (outline)
Flat-target mount (board adapter)	1	SC1-05
Three-ball bed (nest)	1	SC1-06
Run-out fixture: dial indicator on a magnetic base and steel base plate	1	none needed
Rigid sensor mount	1	existing drawing



Section 1b of the procedure; nothing outside sections 1a to 1f is needed for a session.

What must be bought

Item	Purpose	Estimated cost (USD)
Outside micrometer 150-175 mm, with setting standard	Sphere diameter and sphericity, in-house	\$235 to \$300
Calipers with depth rod, 150 mm	Rough TCP length; board distance D; board size	\$30 to \$150
Straightedge 300 mm and feeler gauges	Board flatness	\$75 to \$240
Thermometer	Room temperature at measurement	\$15 to \$40
Torque wrench	Recorded stem and adapter torques	\$80 to \$250
Consumables	Thread locker, retaining compound, shims, alcohol, padded case	\$50 to \$150

A phone camera is enough for the setup photos. Already in hand: sensor, robot, capture computer with its communication software, analysis computer.

Building and buying cost about \$1,875 to \$5,150

Item to build	Estimated cost (USD)
Sphere B with mounting interface	\$500 to \$1,500
Sphere-flange mount: adapter plate and stem	\$200 to \$500
Flat target (board)	\$50 to \$400
Flat-target mount (board adapter)	\$250 to \$600
Three-ball bed (nest)	\$150 to \$450
Run-out fixture: dial indicator, magnetic base, steel base plate	\$240 to \$570
Rigid sensor mount	not estimated

\$1,875 to \$5,150

aluminum sphere B turned by a job shop,
tools included

\$3,375 to \$9,650

sphere B made to order, tools included

Tools and consumables \$485 to \$1,130, included in the totals. Estimates of October 2026, plus or minus 50 percent; sphere B is the least certain.

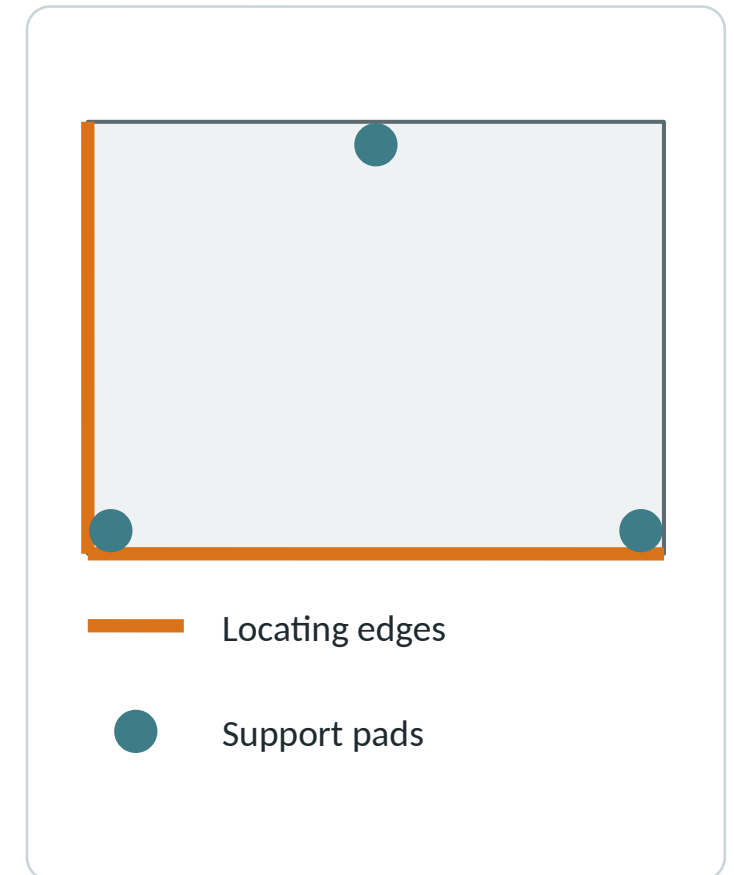
Purchase specification: the target sphere

Requirement	Sphere B
Nominal diameter	152.4 mm (6 inch)
Diameter	Known to 0.025 mm, by micrometer at six or more orientations after finishing, at the session's room temperature
Sphericity	0.02 mm or better
Material	Turned aluminum (6061-T6), solid or hollow with at least 6 mm wall; ceramic as an alternative
Surface	Matte hard anodize, light gray or natural, uniform, free of highlights
Mounting interface	M20 x 30 mm tapped hole at a pole, 32 mm spot face, cut when the sphere is turned
Approximate mass	about 5 kg
Documents	The turner's inspection report: diameter and sphericity as turned, thread and spot face

Procedure section 1d; drawing SC1-04 shows the interface.

Purchase specification: flat plate

Requirement	Flat plate (board)
Size	200 x 150 mm, at least 6 mm thick
Material	Finish-ground aluminum tooling plate (for example MIC-6) or float glass
Flatness	0.05 mm over the front face, checked with a straightedge and a 0.05 mm feeler leaf after painting and after mounting
Front face	Thin, even matte light-gray paint; do not bead-blast a plate thinner than about 10 mm
Edges	Bottom long edge and left short edge square and clean: they rest on the adapter's edge pins
Documents	Supplier's flatness report, kept with the session notes



Procedure section 1d; the board outline is on drawing SC1-05.

The sphere must be matte and measured, not certified



Finish

- Matte hard anodize on the turned aluminum sphere
- A mirror sphere is unusable: one bright highlight
- Steel or ceramic alternative: bead-blast with fine glass bead
- Matte paint is the last resort: 0.02 to 0.05 mm per coat



Diameter

- Known to 0.025 mm, measured after the finish
- Measured in-house by outside micrometer at six or more orientations
- Aluminum grows 0.0035 mm per degree C: measure at session temperature
- The mean goes in the pose log; the spread checks sphericity



Mounting

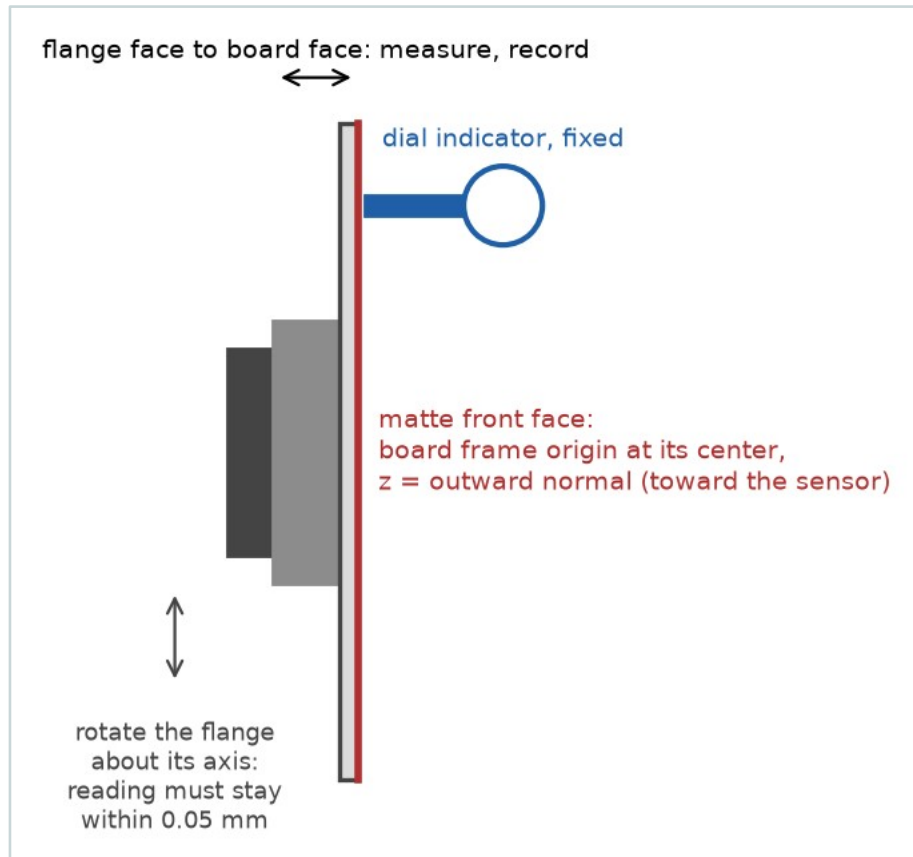
- M20 thread cut at the pole when the sphere is turned
- Hardened steel or ceramic cannot be tapped
- Alternative: blind hole 16 H7 x 46 mm, stem bonded
- No tapped 6 inch precision sphere is stocked

Candidate suppliers

Item	Candidate suppliers	Notes
Sphere, made to order	Precision turning shop (aluminum); Bal-tec; RGP Balls; Industrial Tectonics; Morgan Advanced Materials	A 6 inch matte sphere with a mounting hole is a custom item
Stock reference spheres	Laser-scanner spheres, Rothbucher RSLB10M among them	Not usable: form errors of 0.3 to 2 mm
Board plate	McMaster-Carr MIC-6 plus a grinding shop; Edmund Optics float glass	Painted matte after grinding; flatness checked after mounting
Nest balls	McMaster-Carr; Bal-tec	Grade-25 bearing balls
Adapter, stem, board adapter, nest base	Local machine shop	Inspection report of the toleranced features on each order
Diameter measurement	Outside micrometer 150-175 mm: Mitutoyo 103-143-10, or Starrett	In-house, after the matte finish; setting standard included
Dial indicator and magnetic base	Mitutoyo 2046 series; Noga MG71003 or DG-61003	For the run-out fixture

From suppliers' web pages, October 2026; not an endorsement. Confirm size, finish, measurement and thread before ordering.

Run-out fixture: dial indicator and its mounting



The run-out check: the flange turns, the indicator stays still, the reading must stay within 0.05 mm.



Dial indicator

0.01 mm graduation, about 10 mm travel; for example Mitutoyo 2046 series, about \$50 to \$150



Magnetic base

Articulating, with fine adjustment; for example Noga MG71003 or DG-61003, about \$140 to \$320



Steel base plate

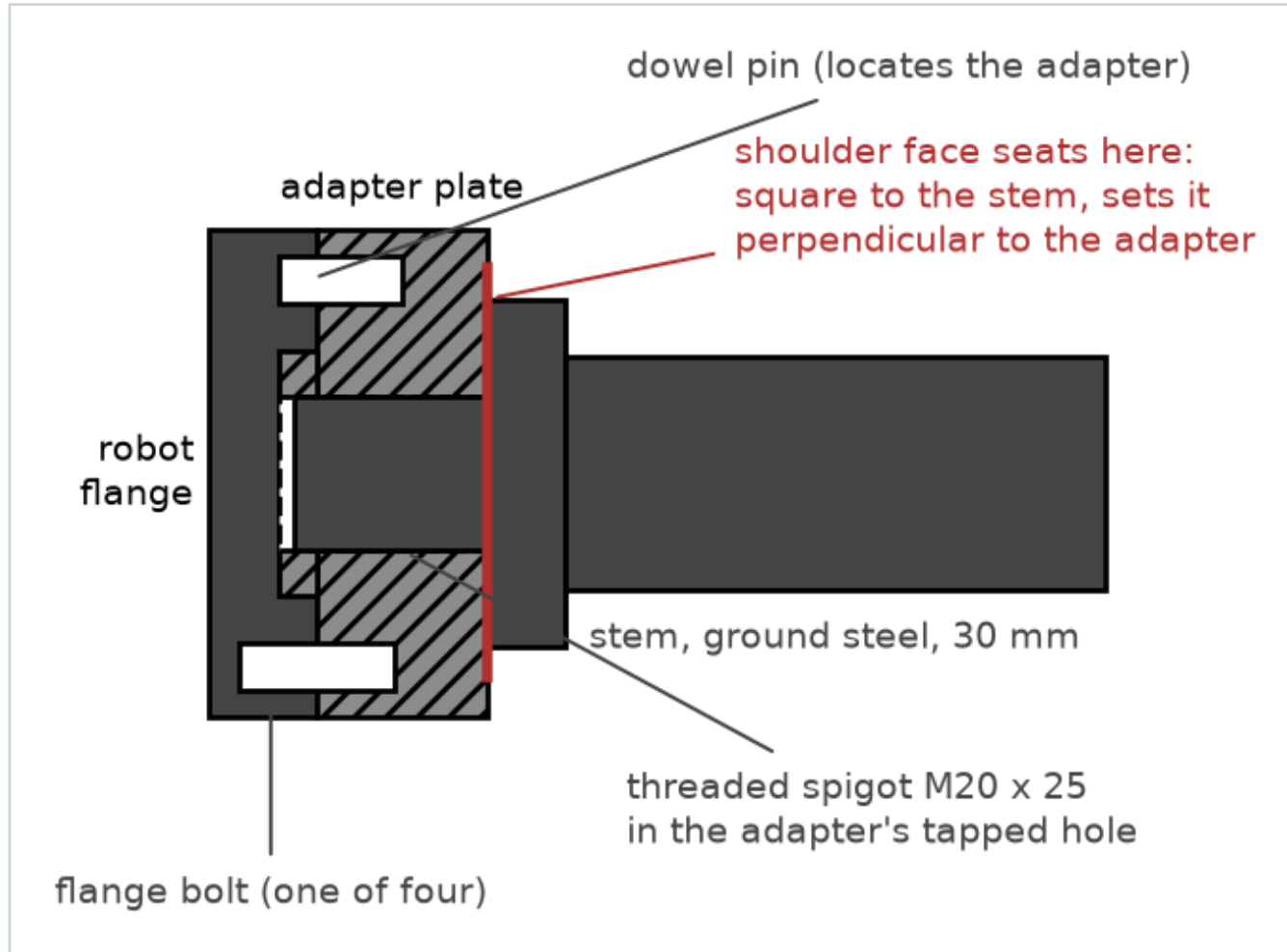
About 150 x 100 x 12 mm, bolted to the cell table with two M8 screws, so the base holds on a non-magnetic table and does not creep



Placement

Where the indicator tip reaches the board's front face about 20 mm from an edge with the robot in its run-out pose

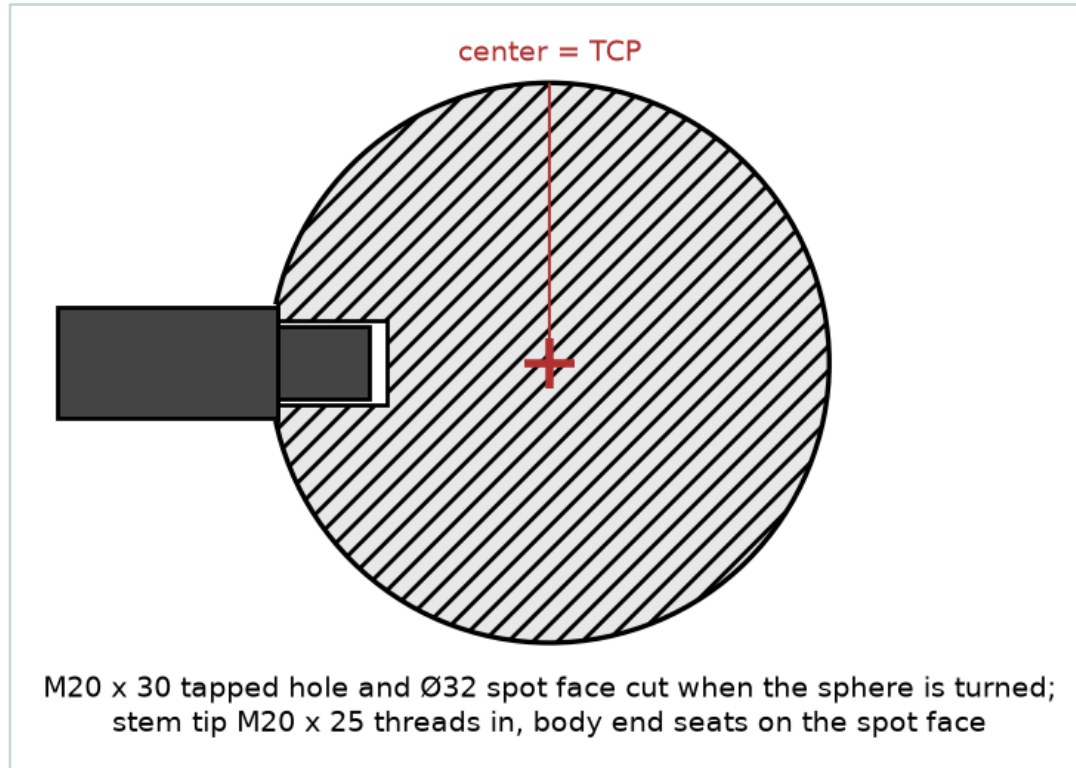
A stiff, repeatable chain from flange to sphere



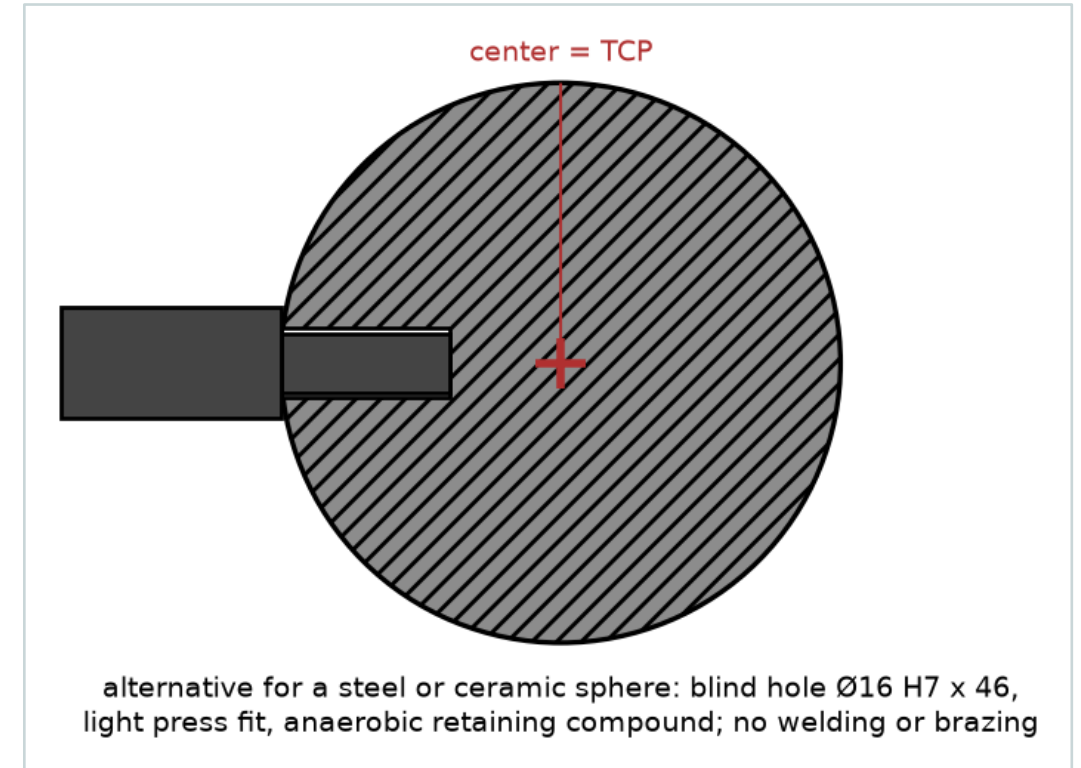
- Adapter located by the flange dowel, so a remount lands in the same place
- Ground 30 mm stem keeps sag to about 0.05 mm
- Stem length at least $2R + 50$ mm, so the sphere hides the stem
- Witness mark across stem and adapter shows a loosened joint
- After any remount, re-check in the nest: readings within 0.1 mm

Adapter on the flange's dowel, with the stem's shoulder seated on the adapter face.

The sphere end: thread cut at the pole, or a bonded stem



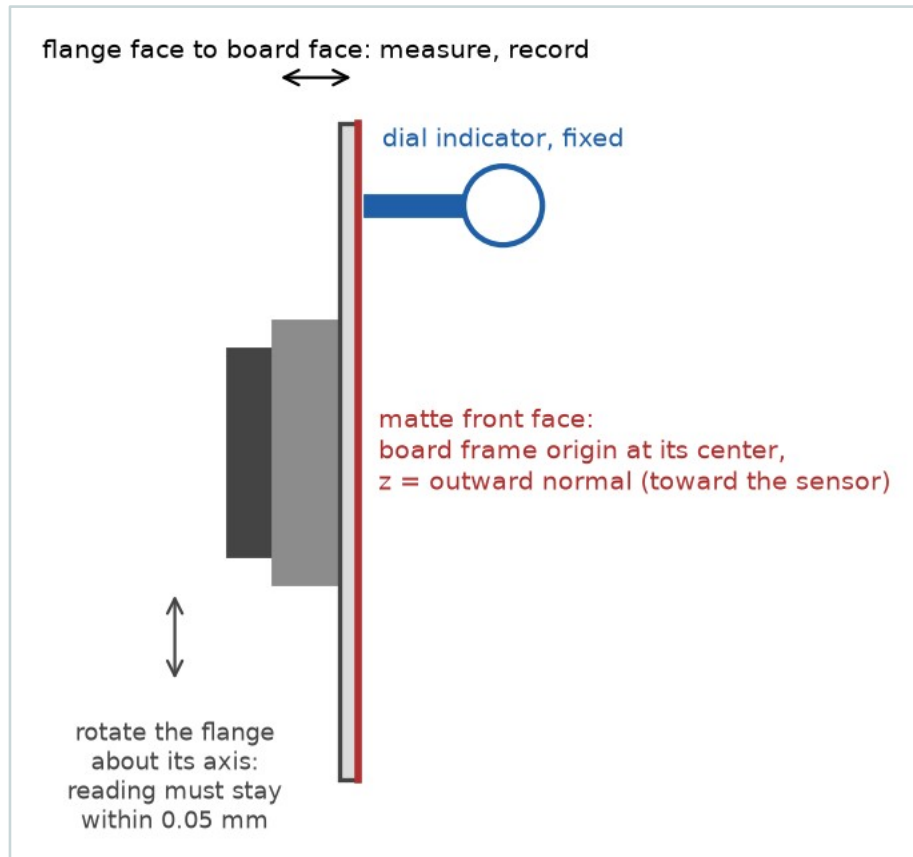
Aluminum sphere B: thread cut at the pole



Steel or ceramic alternative: bonded blind hole

The hole need not be exactly radial: the tool-center-point routine measures the center wherever it ends up.

The board must stay flat after finishing and mounting



Board on its adapter. The runout and frame checks shown here are done at setup (deck 2).



Plate

200 x 150 mm, at least 6 mm thick; ground aluminum tooling plate or float glass



Finish

Matte light gray paint on the front face; do not bead-blast a plate thinner than about 10 mm, it bows



Flatness

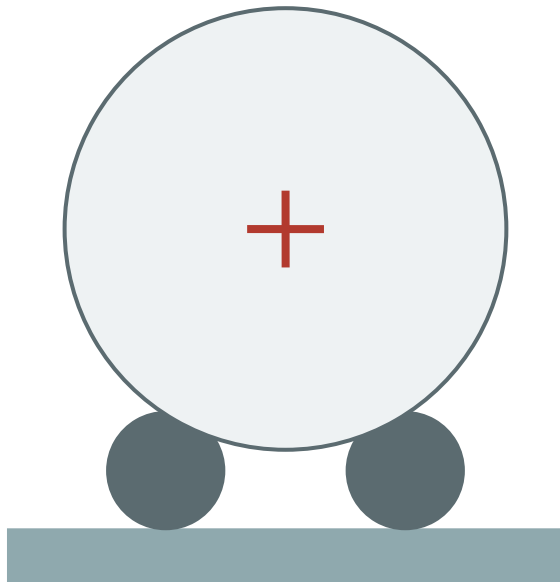
0.05 mm, checked after painting and mounting, not before; keep the flatness report



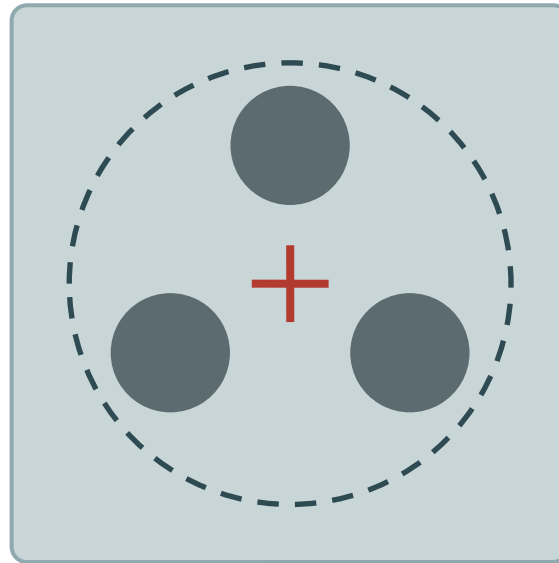
Adapter

Three support pads, three edge pins and three clamp fingers hold the board square to the flange axis, centered on it, and in the same place after a remount

A three-ball nest holds the sphere at a fixed center



Side view



Top view



Three hardened balls, 24 mm diameter, bonded into pockets on a 60 mm circle in a 120 mm steel base



Ordinary grade-25 bearing balls are good enough: the nest's quality comes from the balls being rigidly fixed



Bolted to the table where the robot reaches with the wrist down and tilted about 40 degrees to either side



Used once at setup, and after any remount of the adapter or stem

Checks before the fixtures go to the robot cell

- ✓ Fabricator's inspection report received for the sphere and for SC1-02, SC1-04, SC1-05 and SC1-06
- ✓ Sphere matte, uniform and free of highlights
- ✓ Sphere diameter and sphericity by micrometer after finishing: spread within 0.02 mm, temperature recorded
- ✓ Board flat to 0.05 mm by straightedge and feeler leaf, after painting and mounting
- ✓ Board width and length measured with calipers
- ✓ At setup (deck 2): board run-out within 0.05 mm on the robot; nest re-seat within 0.1 mm
- ✓ Each fixture weighed for its tool load data; witness mark across the stem-adapter joint

